# 2-adic even blow-up Tate normal form `y^2+xy=x^3+a2x^2+a4x+a6` — three-open gluing v33

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**Abstract**

This working v33 manuscript records a three-open `OpenCover` of the *quotient* Rees scheme Proj(ReesMod2) by $D_+(\overline{2t})$, $D_+(\overline{Xt})$, and $D_+(\overline{Yt})$, with affine restriction squares for every ordered pair over abstract product-open pullbacks. `CompatChart2t` (191 lines) identifies the principal divided-chart kernel $(F)$ and proves `chart_eq_quotient_2t`; `CompatChartXt` and `CompatChartYt` (105 lines each) prove $J_X = (G_X : X^\infty)$ and $J_Y = (G_Y : Y^\infty)$ and their quotient-induced chart identifications. Six directed polynomial-to-abstract restrictions satisfy $\mathrm{res}_{\mathrm{poly}} = \mathrm{res}_{\mathrm{actual}} \circ \psi_\bullet$. `SpecialFibreGluing` (167 lines) checks six composed Spec. map compatibilities and the abstract triple cocycle, giving

$$\mathrm{Proj}(\mathrm{ReesMod2}) \cong \mathrm{Glue}(S_{2t}/(2), S_{Xt}/(2), S_{Yt}/(2)).$$

Product overlaps are abstract pullbacks, not polynomial presentations. For $y^2 + xy = x^3 + 8$, the reduced $2t$ chart is $V^2 + UV = 0$; the point $U = V = 0$ lies only in $D_+(\overline{2t})$ (an illustration, not a separately named Lean theorem). `TateEvenBranchGenericFibre` (59 lines) packages the *actual* blow-up comparison $\mathrm{Bl}_I[1/2] \cong \mathrm{Spec}(R_Z[1/2])$ with $I[1/2] = \top$; the existing 2,249-line chart file is untouched. The seven targeted `lake env lean` direct checks previously passed with printed axioms only `propext`, `Classical.choice`, and `Quot.sound`; the scoped `sorry` scan and `git diff --check` are clean. A standalone direct check of the large special-fibre file timed out without a diagnostic and a full `lake build BealGeneral` remains unverified. We do *not* claim a global isomorphism from the actual special-fibre pullback to the quotient Rees Proj, or a degree-one quotient formula or nonvanishing theorem for $\overline{2t}$. This is not a general Beal proof. The intended release and Zenodo source is `beal-conjecture`. The v33 release was cut 2026-09-30 and deposited on Zenodo as DOI 10.5281/zenodo.23054568 (verified).

## 1 Scope and charts

Write $R_Z = \mathbb{Z}_2[X, Y]/(f)$ for the translated even-branch surface under the stated nodal and coefficient-divisibility hypotheses, $I = (2, X, Y)$, and ReesMod2 = Rees$(I)/(2)$, where $(2)$ is generated by the degree-zero base scalar, *not* by $2t$. The integral divided chart has $f(2U, 2V) = 4F$; its principal relation is $(F)$. The $Xt$ and $Yt$ charts use saturated graph ideals $(G_X : X^\infty)$ and $(G_Y : Y^\infty)$ rather than the unsaturated graph equations. Quotient-induced chart maps connect these integral calculations to the three reduced chart spectra. These assertions concern the charts under their hypotheses, not an unconditional formula for every Weierstrass model.

## 2 The checked gluing boundary

The three reduced generator opens cover Proj(ReesMod2). Ordered affine restriction squares live over product opens; the six distinct directed pairs obey the polynomial-to-abstract restriction

identities. The polynomial spectra are literal cover objects, but their *overlaps* are abstract product pullbacks. Six Spec. map squares and a triple cocycle yield the global isomorphism displayed in the abstract. No polynomial ring presentation of a product overlap is supplied or inferred.

The illustrative $y^2 + xy = x^3 + 8$ specialization has $f(2U, 2V) = 4(V^2 + UV - 2U^3 - 2)$, so its reduced divided-chart equation is $V^2 + UV = 0$. The point $(U, V) = (0, 0)$ gives direction $[2t : Xt : Yt] = [1 : 0 : 0]$ and belongs to the $2t$ open alone; this is an example, not a global classification theorem.

## 3 Generic fibre and unresolved special-fibre comparison

After inverting 2, the centre $I$ becomes the unit ideal. The packaged generic-fibre isomorphism concerns the *actual* blow-up and the localized translated surface over $D(2)$; it does not identify an integral chart with its generic fibre. In contrast, the checked special-fibre gluing concerns the *quotient Rees Proj*. Chartwise base-change comparisons do not yet give an isomorphism of that Proj with the actual scheme-theoretic special-fibre pullback. Nor is $(\mathrm{ReesMod2})_1 \cong (I/2I)t$ or $\overline{2t} \neq 0$ certified here. These missing steps are not used as premises of the quotient-Proj gluing.

## 4 Verification and deposit route

The seven direct Lean checks and their axiom reports are recorded in `Beal/RELEASE_NOTES_v33.md` and `Beal/AUDIT_v33_HONESTY.md`. They are not a verified full-target build: the standalone large special-fibre source check timed out, although the passing modules imported its compiled declarations. The existing `TateEvenBranch.lean` (2,249 lines) passed its separate direct check and was not refactored.

`beal-conjecture` is the public repository and the intended GitHub-release source. The level-26 scaffold in this tree is not an inhabited global Beal result. An authenticated GitHub-side check found one active Zenodo hook subscribed to `release` events for the main repository. v33 was released 2026-09-30; the Zenodo record, archive, metadata, and DOI 10.5281/zenodo.23054568 were verified after release. No mirror push is authorized by this manuscript.